FLUID STATICS

Pressure:

- Pressure is defined as

$$P = \frac{F}{A}$$

F is the force of impact of molecules on the walls Newtons

A is the area of the walls m^2

The unit for pressure in the SI system is the Pa (Pascal) $=1N/m^2$

However, there are some other pressure-units, which are still in use: bar, atm., and Torr(1mm hg).

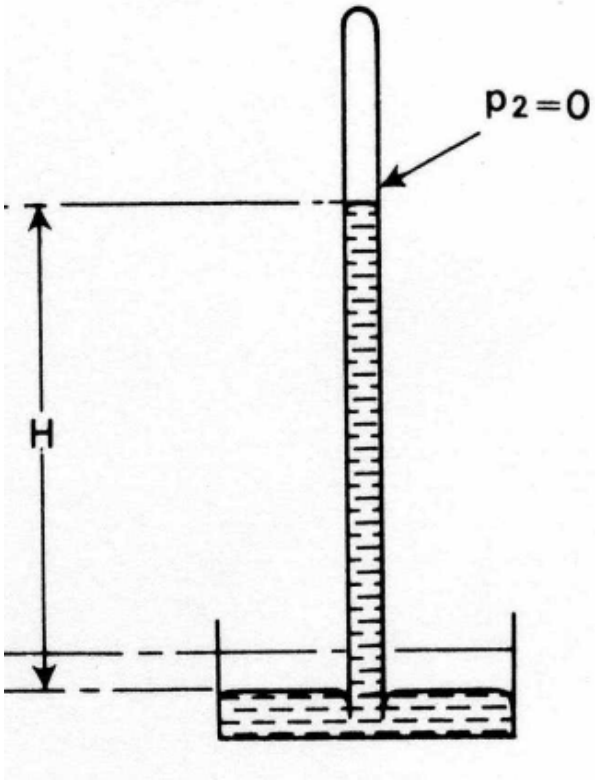
For example, the pressure at sea level is 101300 Pa at standard temperature and the most common name for this specific pressure is an atmosphere or atm.

The closest match to the atmosphere is 1 bar = $1x10^5$Pa

The ratios between different units are given in Table

Pressure Unit	Pa	Bar	Atm	Torr
Pa	1	0.00001	9.869×10^{-6}	7.501×10^{-3}
Bar	100000	1	9.869×10^{-1}	750.1
Atm	101325	1.01325	1	760
Torr	113.22	0.001333	1.316×10^{-3}	1

Pressure can also referred as column of liguid.
Atm pressure= 760 mm Hg
Barometer



Pascal's law

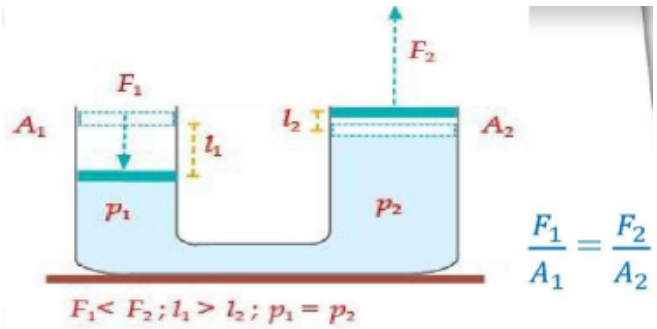
Blaise Pascal

1653

Pascal's law states that the pressure or intensity of pressure at a point in a static fluid(at rest) is equal in all directions.

Applications

- Hydraulic lift
- Hydraulic Jacks
- Hydraulic Brakes in vehicles
- Hydraulic Pumps



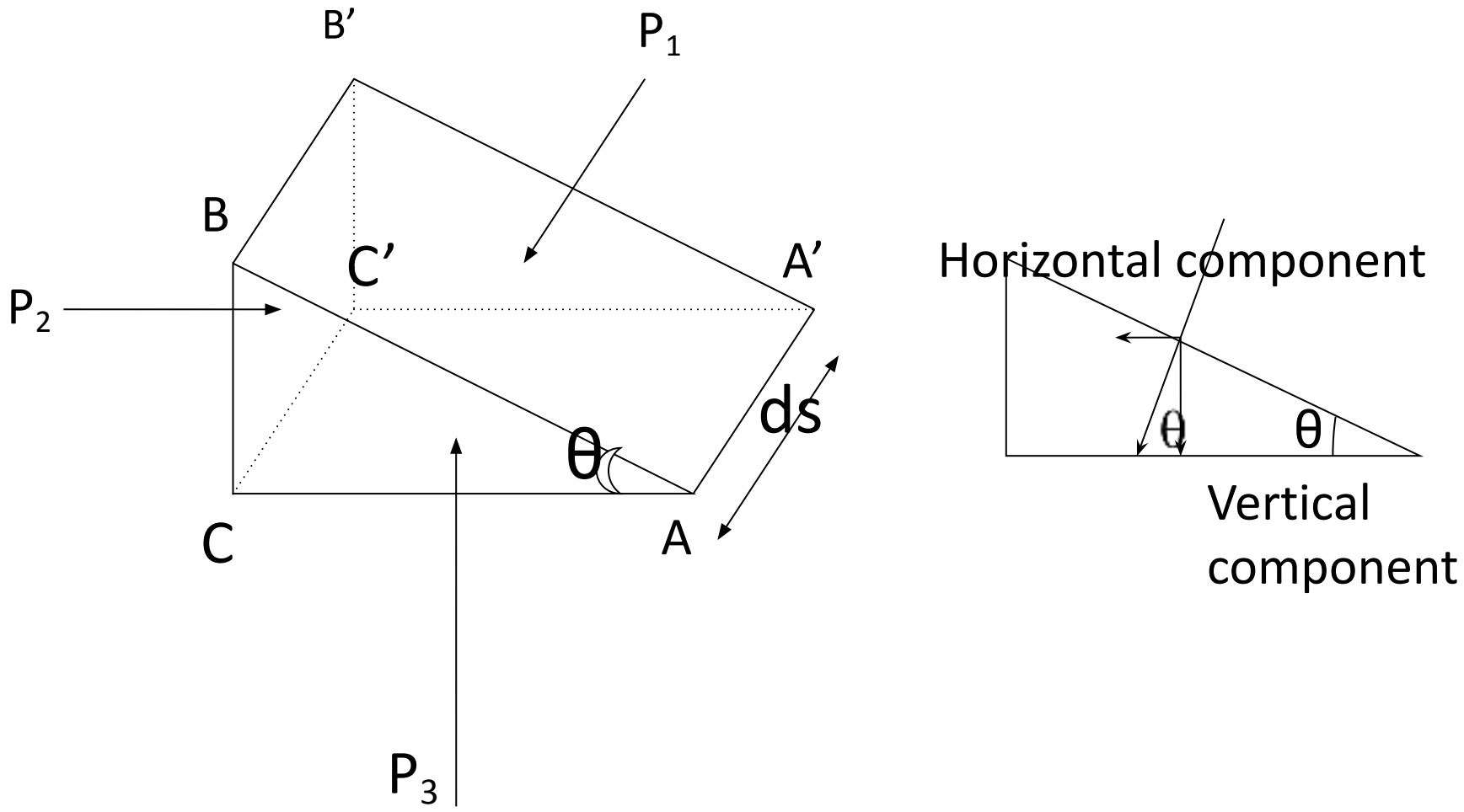


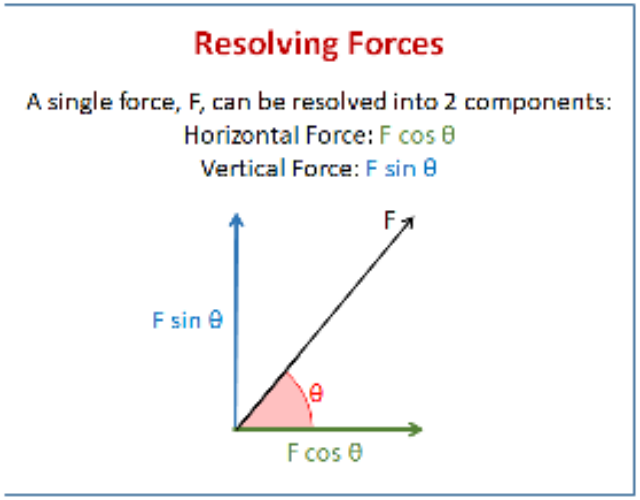
Figure : Forces acting on a small element

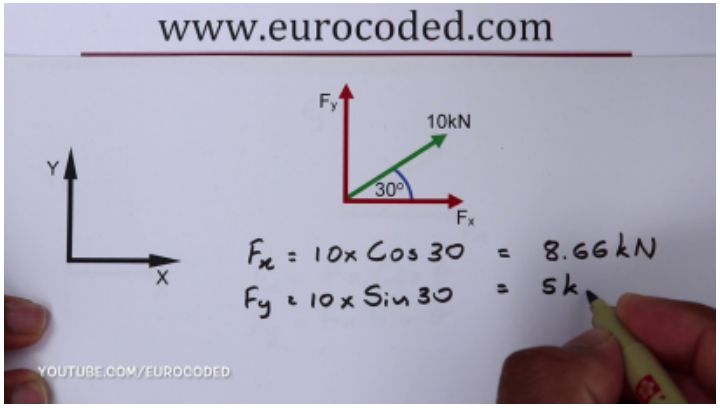
Proof: Consider a very small prism (wedge shaped element) in the fluid at rest and assume that its thickness is ds=1.

Let the pressures in three directions of the prism are P1, P2 and P3.

Under the action of the pressure forces the prism is in equilibrium and therefore the forces in horizontal and vertical directions must be balanced.

Since a very small element is considered the gravitational fo





let the angle BAC $= \theta$

Resolving the forces in the horizontal direction

$P_2 .BC – P_1 . AB . \sin\theta = 0 \rightarrow (1)$

$$\sin\theta = \frac{BC}{AB}$$

Substituting the value of $\sin\theta$ in equation (1)

$P_2 .BC. – P_1 . AB.(BC/AB). = 0$

$\Rightarrow P_2 = P_1$

Resolving the forces in vertical direction

Since the element is very small the force due to its weight can be neglected and the above equation can be written as

$$P_3 \cdot AC - P_1 \cdot AB \cdot \cos\theta = 0 \rightarrow (2)$$

From the Figure $\cos\theta = \frac{AC}{AB}$

substituting the value of cos θ in the equation 2,

$$P_3 \cdot AC - P_1 \cdot AB \cdot (AC/AB) = 0 \rightarrow$$

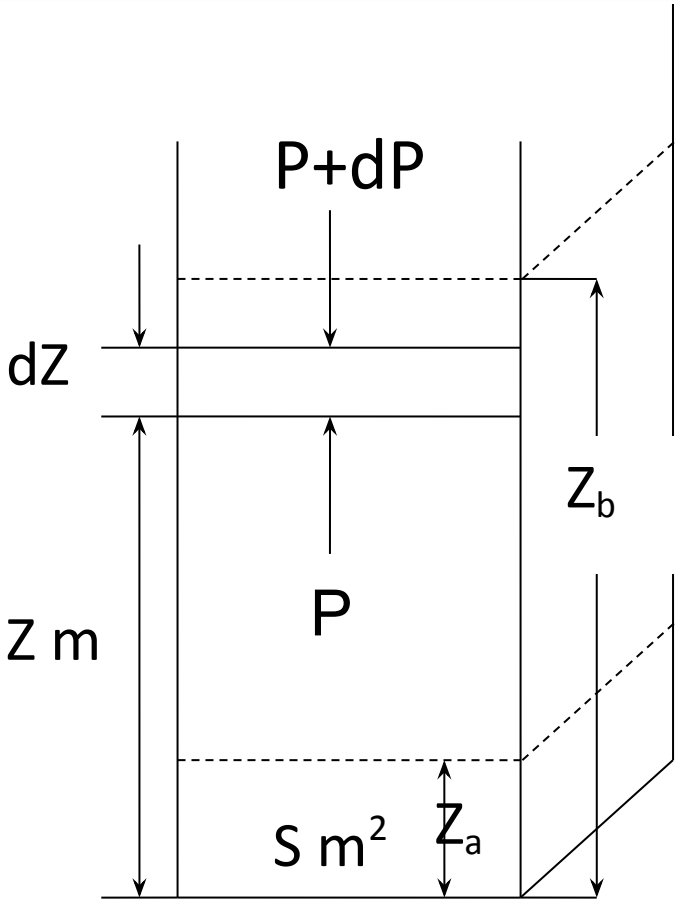
$$P_3 = P_1$$

$$\therefore P_1 = P_2 = P_3$$

From this it can be concluded that the pressure at a point in a static fluid is same in all directions.

Pressure variations with height in static fluids

Consider a vertical column of fluid as shown in figure



Let the cross-sectional area of the column be ‘S’ m^2
At a height ‘Z’ m from floor let the pressure be P and let (P+dP) be the pressure at (Z+dZ) m.
Assume the density of the fluid be ‘ρ’ kg/ m^3.
To find the pressure variations along the vertical line write the force balance around
the volume S dZ.
The three vertical forces acting on this are
The force of gravity acting downward is ρ.S.dZ. g

The force from pressure ‘P’ acting in the upward direction is P.S
The force from pressure ‘P+dP’ acting in downward direction is (P+dP). S
When the fluid is in equilibrium sum of the three forces must be equal to zero.

P.S – (P+dP).S- ρ . S. dZ.g = 0

Divide through out by ‘S’

P – P – dP - ρ. dZ. g = 0 ;

$$\frac{dP}{\rho} = -g\,.\,dZ \qquad \text{----------(1)}$$

The above equation can be integrated BASED ON variation of density with pressure.

$$\int_{pa}^{pb} dp = \int_{za}^{zb} dz$$

Case i : Assuming density is constant with respect to pressure ie. the fluid is incompressible. (Hydrostatic Equilibrium)

$$\frac{dP}{\rho} + g.dZ = 0$$

Integrating this equation gives, $\frac{P}{\rho} + g.Z = \text{constan}$

The same equation can be integrated between the limits Z_a and Z_b

$$\frac{P_b - P_a}{\rho} = g(Z_a - Z_b)$$

So, the pressure varies along the vertical line. As the depth increases pressure will also increase.

The above equation can be written as $\Delta P = \rho\, g\, \Delta\, h$, $\Delta\, h$ is the height measured form the free surface.

This equation is called hydrostatic equation.

The pressure imposed by the height of water is called hydrostatic pressure and is directly proportional to the height of the fluid above and its density. The mathematical definition of hydrostatic pressure is

$$P = \rho g z$$

where z is the height of the fluid, ρ is fluid density and g is gravitational acceleration.

Case ii. Assuming density is varying with respect to pressure (gases).

Barometric Equation

Case a: Assuming the fluid obeys isothermal law.(PV=nRT)

For isothermal process, the density and pressure are related by the equation

$$\rho = \frac{PM}{RT}$$

$\rho = m/v$

P is pressure in N/m^2

M is molecular weight of the fluid in kg/kgmole=n/m

T is absolute temperature in K and

R is universal gas constant = 8314.4 J/(kgmole K)

Consider equation (1)

$$\frac{dP}{\rho} + g.dZ = 0$$

Substituting the value of ρ in the above equation, we have

$$\frac{dP}{P} + \frac{gM}{RT} dZ = 0$$

Integrating the above equation between the levels a and b on the assumption that T is constant, gives

$$\ln \frac{P_b}{P_a} = -\frac{gM}{RT}[Z_b - Z_a]$$

$$\frac{P_b}{P_a} = e^{-\frac{gM}{RT}[Z_b - Z_a]}$$

This equation is called barometric equation

$$\rho = \frac{PM}{RT}$$

This can be written as $\frac{M}{RT_a} = \frac{\rho_a}{P_a}$

Therefore barometric equation can be written as $\frac{P_b}{P_a} = e^{-\frac{g\rho_a}{P_a}[Z_b - Z_a]}$

Or

$$P_b = P_a e^{-\frac{g\rho_a}{P_a}[Z_b - Z_a]}$$

Case b : Assuming the fluid obeys Adiabatic law.

For adiabatic process the relation between pressure and density is given by

$$\frac{P}{\rho^k} = \text{constant} = c \rightarrow (2)$$

where k is ratio of specific heat constants ie., C_p/C_v.

$$\rho^k = \frac{P}{c} \Rightarrow \rho = \left(\frac{P}{c}\right)^{1/k}$$

we know from equation (1)

$$\frac{dP}{\rho} + g.dZ = 0$$

substituting the value of ρ in the above equation gives

$$\frac{dP}{\left(P/c\right)^{1/k}} = -g.dZ \quad \Rightarrow \quad \frac{c^{1/k}\,dP}{P^{1/k}} = -g.dZ$$

Integrating the above equation gives

$$c^{1/k}\int_{P_a}^{P_b}\frac{dP}{P^{1/k}} = -g\int_{Z_a}^{Z_b}dZ$$

$$\left[c^{1/K} \frac{P^{-\frac{1}{k}+1}}{-\frac{1}{k}+1} \right]_{P_a}^{P_b} = -gZ\Big|_{Z_z}^{Z_b}$$

from equation (2)

$$c^{1/K} = \frac{P^{1/K}}{\rho}$$

Substituting for '$c^{1/K}$' in the above equation gives

$$\left[\frac{P}{\rho}^{1/K} * \frac{P^{-\frac{1}{k}+1}}{-\frac{1}{k}+1} \right]_{P_a}^{P_b} = -gZ\Big|_{Z_a}^{Z_b}$$

$$\left.\frac{P^{1/K}\,P^{\frac{K-1}{K}}}{\rho\,\frac{K-1}{K}}\right|_{P_a}^{P_b} = -g[Z_b - Z_a]$$

$$\left.\frac{K}{K-1}\left(\frac{P}{\rho}\right)\right|_{P_a}^{P_b} = -g[Z_b - Z_a]$$

Rearrangin
g

$$\frac{K}{K-1}\left(\frac{P_b}{\rho_b}-\frac{P_a}{\rho_a}\right)=-g[Z_b-Z_a]$$

Expression to find temperature lapse rate

$$\left(\frac{P_b}{\rho_b}\right)=\frac{P_a}{\rho_a}-\left(\frac{K-1}{K}\right)g[Z_b-Z_a]$$

Multiplying both sides by ρ_a/P_a, gives

$$\left(\frac{P_b}{\rho_b}\right)\left(\frac{\rho_a}{P_a}\right)=1-\left(\frac{K-1}{K}\right)g[Z_b-Z_a]\left(\frac{\rho_a}{P_a}\right) \quad \text{-------(3)}$$

From equation (2) we have

$$\frac{P}{\rho^k} = \text{constant} = c \rightarrow (2)$$

$$\frac{P_a}{\rho_a^{\,k}} = \text{constant} = c \text{- - - - - } (2a)$$

$$\frac{P_b}{\rho_b^{\,k}} = \text{constant} = c \text{- - - - } (2b)$$

Dividing 2a by 2b

$$\frac{P_a}{P_b} = \left(\frac{\rho_a}{\rho_b}\right)^k$$

$$or \left(\frac{\rho_a}{\rho_b} \right) = \left(\frac{P_a}{P_b} \right)^{1/k}$$

Substituting for ρ_a/ρ_b in equation 3 gives

$$\left(\frac{P_b}{P_a} \right) \left(\frac{P_a}{P_b} \right)^{1/k} = 1 - \left(\frac{K - 1}{K} \right) g[Z_b - Z_a] \left(\frac{\rho_a}{P_a} \right)$$

$$\left(\frac{P_b}{P_a} \right) \left(\frac{P_b}{P_a} \right)^{-(1/k)} = 1 - \left(\frac{K - 1}{K} \right) g[Z_b - Z_a] \left(\frac{\rho_a}{P_a} \right)$$

$$\left(\frac{P_b}{P_a}\right)^{\frac{k-1}{k}} = 1 - \left(\frac{K-1}{K}\right) g[Z_b - Z_a]\left(\frac{\rho_a}{P_a}\right)$$

or

$$\left(\frac{P_b}{P_a}\right) = \left[1 - \left(\frac{K-1}{K}\right) g[Z_b - Z_a]\left(\frac{\rho_a}{P_a}\right)\right]^{\frac{K}{K-1}} \quad - - - \; 4$$

Pressure at a height Z_b m from Z_a is given by

$$P_b = P_a\left[1 - \left(\frac{K-1}{K}\right) g[Z_b - Z_a]\left(\frac{\rho_a}{P_a}\right)\right]^{\frac{K}{K-1}}$$

If datum is taken as $Z_a = 0$, where temperature, pressure and density are T_o, P_o and ρ_o
Then the above equation becomes

$$P_b = P_o\left[1 - \left(\frac{K-1}{K}\right) g Z_b \left(\frac{\rho_o}{P_o}\right)\right]^{\frac{K}{K-1}} \quad - - - - - 5$$

From ideal gas law

$$\frac{P_o}{\rho_o} = \frac{RT_o}{M}$$

$$P_b = P_o\left[1 - \left(\frac{K-1}{K}\right) gZ_b\left(\frac{\rho_o}{P_o}\right)\right]^{\frac{K}{K-1}}$$

$$P_b = P_o\left[1 - \left(\frac{K-1}{K}\right)\left(\frac{gMZb}{RT_o}\right)\right]^{\frac{K}{K-1}} \quad - - - - - 6$$

Inclusion of the variation of pressure withtemperature

To find temperature at a height

Z_b

We have from ideal gas law,

$$\rho_b = \frac{P_b M}{RT_b} \quad \text{And} \quad \rho_o = \frac{PoM}{RT_o}$$

$$\frac{T_b}{T_0} = \frac{P_b \rho_o}{P_o \rho_{b_1}} \qquad or \left(\frac{\rho_o}{\rho_b}\right) = \left(\frac{P_o}{P_b}\right)^{1/k}$$

$$\Rightarrow \frac{T_b}{T_0} = \frac{P_b}{P_o}\left(\frac{P_b}{P_o}\right)^{-\frac{1}{K}}$$

$$\Rightarrow \frac{T_b}{T_0} = \left(\frac{P_b}{P_o}\right)^{\frac{k-1}{k}}$$

Substituting value of P_b/P_o from equation 4 gives

$$\left(\frac{P_b}{P_o}\right) = \left[1 - \left(\frac{K-1}{K}\right) g[Z_o - Z_b]\left(\frac{\rho_o}{P_o}\right)\right]^{\frac{K}{K-1}} \quad \text{- - - 4}$$

$$\Rightarrow \frac{T_b}{T_0} = \left(\left[1 - \left(\frac{K-1}{K}\right) g Z_b \left(\frac{\rho_o}{P_o}\right)\right]^{\frac{K}{K-1}}\right)^{\frac{K-1}{K}}$$

$$\frac{T_b}{T_0} = \left[1 - \left(\frac{K-1}{K}\right) g Z_b \left(\frac{\rho_o}{P_o}\right)\right]$$

$$T_b = T_0\left[1 - \left(\frac{K-1}{K}\right) g Z_b \left(\frac{\rho_o}{P_o}\right)\right] \quad \text{----------(7)}$$

Expression(7) to find Temperature at any height Zb based on the reference position

Temperature lapse rate: It is defined as the rate at which the temperature changes with elevation.

Equation 7 is modified by replacing

$$\frac{P_o}{\rho_o} = \frac{RT_o}{M}$$

$$T_b = T_0\left[1 - \left(\frac{K-1}{K}\right)\frac{gMZ_b}{RT_o}\right]$$

To obtain an expression for the temperature lapse rate, the temperature given by the above equation is differentiated with respect to Z_b

$$\frac{dT_b}{dZ_b} = \frac{d}{dZ_b}\left\{T_o\left[1 - \left(\frac{K - 1}{K}\right)\left(\frac{gMZ_b}{RT_o}\right)\right]\right\}$$

$$\frac{dT_b}{dZ_b} = -\left(\frac{K - 1}{K}\right)\left(\frac{gM}{RT_o}\right)T_o$$

$$\frac{dT_b}{dZ_b} = L = -\left(\frac{gM}{R}\right)\left(\frac{K-1}{K}\right) \quad \ldots\ldots\ldots\ldots 8$$

Note : 1. If ‘K’ is not given take, K = 1.4, at ideal conditions.

2. R = 8314.4 J/kgmole K

3. If adiabatic process is reversible it is known as Isentropic process.If in any problem if it is stated that it is isentropic process equation 8 can be used

1. Find the head 'h' of water corresponding to an intensity of pressure of 981KN/m^2. Density of water is 1000kg/m^3.

Data : P = 981*10^3 N/m^2.

ρ = 1000 kg/m^3.

h = ?

we have P = ρ g h

$$\Rightarrow \quad h = \frac{P}{\rho g} = \frac{981*10^3}{1000*9.81} = 100m$$

2. The pressure intensity at a point in a fluid is given 39.24 kN/m^2. Find the corresponding height of fluid, when the fluid is a) water and b) oil of specific gravity 0.9.

Data : (i) P = 39.24 kN/m^2 = $39.24 * 10^3$ N/m^2

ρ = 1000 kg/m^3.

h = ?

(ii) P = 39.24 kN/m^2 = $39.24 * 10^3$ N/m^2

ρ = 900 kg/m^3.

h = ?

(i) $$h = \frac{P}{\rho g} = \frac{39.24 * 10^3}{1000 * 9.81} = 4m$$

(ii) $$h = \frac{P}{\rho g} = \frac{39.24 * 10^3}{900 * 9.81} = 4.44m$$

3. If the atmosphere pressure at sea level is 10.143 N/cm^2, determine the pressure at a height of 2500m assuming the pressure variation follows hydrostatic law and isothermal law. The density of air is given as 1.208kg/m^3.

Data : $P_a = 10.143\ N/cm^2 = 10.143 * 10^4\ N/m^2$

$z_b = 2500m$, $z_a = 0$, $\rho_a = 1.208\ kg/m^3$

- assuming hydrostatic law applies,

$$\int_{P_a}^{P_b} \frac{dp}{\rho} = -\, g \int_{z_a}^{z_b} dz$$

$(P_b - P_a) = - \rho g (z_b - z_a)$

$P_b = P_a - \rho g(z_b - z_a) = 10.143*10^4 - 1.208*9.81*2500$

$= 71803.8 \text{ N/m}^2$

ii) assuming Isothermal law applies,

$\mathbf{P_b = P_a * exp(-g\rho_a (z_b - z_a)/P_a)}$

$\mathbf{= 10.143*10^4 * exp(9.81*1.208*2500)/(10.143*10^4))}$

$\mathbf{= 75738.394 \text{ N/m}^2}$

The atmospheric pressure at the sea level is 101.3 KN/m^2 and temperature is 15°C. Calculate the pressure at 8000m above seal level, assuming (i) isothermal variation of density, and (ii) adiabatic variation of pressure and density. Assume density of air as sea-level as 1.285kg/m^3. Neglect variation of 'g' with altitude.

Data : $P_a = 101.3\ kN/m^2 = 101.3 * 10^3\ N/m^2$

$z_b = 8000m$, $z_a = 0$, $\rho_a = 1.285\ kg/m^3$

.

(i) *assuming Isothermal law applies,*

$$P_b = P_a * \exp(-g\rho_a\ (z_b - za)/P_a)$$

$$= 101.3*10^3 * \exp(-(9.81*1.285*8000)/(1.013*10^5))$$

$$= 37433.28\ N/m^2$$

(ii) From equation 4

Assuming K = 1.4

$$P_b = P_a\left[1 - \left(\frac{K-1}{K}\right)\frac{gz_b\rho_a}{P_a}\right]^{\frac{K}{K-1}}$$

$$P_b = 101300\left[1 - \left(\frac{1.4-1}{1.4}\right)\frac{9.81*8000*1.285}{101300}\right]^{\frac{1.4}{1.4-1}}$$

$$= 31396.341\ N/m^2$$

The temperature of earth's atmosphere drops about 5°C for every 1000 m of elevation above the earth's surface. If the air temperature at ground level is 15°C and the pressure is 760 mm Hg, at what elevation is the pressure 380 mm Hg? Assume that the air behave as an ideal gas.

Data : P_a = 760 mm Hg = 101325 N/m^2 , L = -0.005°C/m

P_b = 380 mm Hg = (380/760)*101325 = 50662.5 N/m^2

z_b = ? m, z_a = 0, ρ_a = 1.285 kg/m^3, T_a = 15°C = 273+15 = 288 K.

for every 1000 m the decrease in temperature is 5 °C

∴for 1m the decrease in temperature is 0.005° C

$$L = -\frac{gM}{R}\left(\frac{K-1}{K}\right)$$

$$-0.005 = -\frac{9.81*29}{83144}\left(\frac{K-1}{K}\right)$$

$$K = 1.17$$

$$\frac{1}{K}$$

$$P_b = P_a\left[1-\left(\frac{K-1}{K}\right)\frac{gz_bM}{RT_a}\right]^{\frac{K}{K-1}}$$

$$506625 = 1013251-\left(\frac{1.171-1}{1.171}\right)\frac{9.81*Z*29}{83144*288}\Bigg]^{\frac{1.171}{1.171-1}}$$

$$Z = 5553.778\text{m}$$

Alternate method

$$\frac{dP}{P} = - \frac{gM}{RT} dZ$$

Drop in temperature = 5 deg for 1000 m height

Drop in temperature for Z m = 5 Z/1000 deg

Temperature any height = 288 - 5z/1000 deg K

=288-0.005Z

Substituting for T

$$\frac{dP}{P} = - \frac{gM}{R(T - 0.005Z)} dZ$$

Integrating with limits

$$\int_{P1}^{P2} \frac{dP}{P} = - \int_{0}^{Z} \frac{gM}{R(T - 0.005Z)} dZ$$

$$\int_{P1}^{P2} \frac{dP}{P} = - \frac{gM}{R} \int_{0}^{Z} \frac{dZ}{(T - 0.005Z)}$$

P1=760mmHg,
P2=380 mmHg
T=288

$= - \ln(a-bx)$

$$= - \quad \text{z} \quad 0$$